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Training in the evaluation of pesticides (plant protection products and active substances) according to Regulation (EC) No 1107/2009

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Abstract

Plant Protection Products, also called 'pesticides', are intended to protect crops by controlling pests, weeds and diseases. This Technical Report contains a description of the activities within the work programme of the EU-FORA Fellowship on the training in the evaluation of pesticides (active substances and plant protection products) according to Regulation (EC) No 1107/2009. The purpose of the Plant Protection Product Regulation is to ensure a high level of protection of both human and animal health and the environment and at the same time to improve the functioning of the internal market through the harmonisation of the rules on the placing on the market of plant protection products, while improving agricultural production. The data requirements for active substances and plant protection products are set out in Regulations (EC) 283/2013 and 284/2013, covering active substance(s), safeners or synergists. The scope of this work programme was to help the fellow to gain knowledge and experience in order to understand the evaluation process of the data submitted by the Applicants at each different section of an active substance and a plant protection product dossier in a regulatory and scientific basis.

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Summary

This Technical Report focuses on the activities within the work programme of the EU-FORA Fellowship on the training in the evaluation of pesticides (active substances and plant protection products) according to Regulation (EC) No 1107/2009. All the relevant areas of evaluation of an active substance and plant protection product dossier according to Regulations (EC) 1107/2009, 283/2013, 284/2013 have been considered for the training of the fellow student. More specifically, identity, physicochemical properties, methods of analysis, efficacy, residues, mammalian toxicology, fate and behaviour in the environment, ecotoxicology and classification and labelling were discussed through the prism of the existing regulatory framework. Furthermore, the requirements and the EU common approaches, have been scientifically analysed in each section. An introduction in specific requirements (legislation framework – data requirements) for each section, followed by hands-on-training (specific case studies), have been performed.

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1. Introduction

Plant protection products (PPPs), also referred as pesticides, contain at least one active substance. These active substances can be chemicals, micro-organisms, pheromones, botanical extracts or viruses that enable PPPs to perform a controlling effect on the target pests, diseases or weeds. They may also contain other components such as co-formulants, safeners and synergists. The active substance(s), safeners or synergists, are intended for one of the following uses:

- protecting plants or plant products against harmful organisms or preventing the action of such organisms, unless the main purpose of these products is considered to be for reasons of hygiene rather than for the protection of plants or plant products (e.g. fungicides, insecticides);
- influencing the life processes of plants, such as substances influencing their growth, other than as a nutrient (e.g. plant growth regulators, rooting hormones);
- preserving plant products, in so far as such substances or products are not subject to special Community provisions on preservatives (e.g. extending the life of cut flowers);
- destroying undesired plants or parts of plants, except algae unless the products are applied on soil or water to protect plants (e.g. herbicides/weedkillers to kill actively growing weeds);
- checking or preventing undesired growth of plants, except algae unless the products are applied on soil or water to protect plants (e.g. herbicides/weedkillers preventing the growth of weeds).

As an example, PPPs may include insecticides, fungicides, herbicides, molluscicides and plant growth regulators.

Regulation (EC) No 1107/2009 is the main legislative framework regulating the placing on the market of PPPs. The Regulation lays down the rules and procedures for the approval of active substances and the authorisation of PPPs. For active substances to be used in PPPs, they must fulfil the approval criteria while PPPs cannot be placed on the market or used without prior authorisation in an EU Member State.

This specific project '*Training in the evaluation of pesticides (plant protection products and active substances) according to Regulation (EC) No 1107/2009*' in the context of the EU-FORA fellowship programme was implemented by the hosting site Benaki Phytopathological Institute (BPI), which specialises in risk assessment and advice on the safe usage of agricultural chemicals with regards to the protection of human health and that of the environment.

2. Description of the work programme

2.1. Aims

The training programme followed the evaluation of pesticides according to the European Legislation framework and the risk assessment of PPPs and their active substances in the areas of identity and physicochemical properties, methods of analysis, efficacy, residues, mammalian toxicology, fate and behaviour in the environment and ecotoxicology, as well as classification and labelling. As an introduction the training focused on the general aspects of evaluation procedures, existing Legislation framework including guidance documents related to horizontal procedures of the evaluation of PPPs. Then, a comprehensive presentation of the various evaluation sections including evaluated case studies was delivered by key experts in order to obtain a first overview of the evaluation procedure and the interconnections between the different sections of PPPs evaluation dossiers.

2.2. European legislation framework

Regulation (EC) No 1107/2009¹ is the legislation concerning the placing of PPPs on the market in the European Union. This Regulation specifies the cut-off criteria for the approval of active substances and decision-making criteria for the authorisation of PPPs at Member State level. In accordance with the mentioned Regulation, approval of active substances in PPPs including the definition of maximum residue levels is coordinated at the European level with involvement of the European Food Safety

¹ Regulation (EC) No 1107/2009 of the European Parliament and of the Council of 21 October 2009 concerning the placing of plant protection products on the market and repealing Council Directives 79/117/EEC and 91/414/EEC. OJ L 309, 24.11.2009, p. 1–50.

Authority (EFSA). EFSA is responsible for the EU peer review of risk assessments of active substances used in PPPs, in close cooperation with EU Member States and the European Commission. However, the authorisation procedure for PPPs such as zonal and mutual recognition as well as additional applications of authorised PPPs are coordinated by Member States.

The data requirements for the active substance provided for in Article 8(1)(b) of Regulation (EC) No 1107/2009 are set out in the Annex to Regulation (EU) No 283/2013,² The respective data requirements for a PPP for in Article 8(1)(b) of Regulation (EC) No 1107/2009 are set out in the Annex to Regulation (EU) No 284/2013³.

The evaluation and authorisation of PPPs is also subject to Uniform Principles in accordance with Regulation (EU) No 546/2011.⁴ These principles include assessment of impact on animal and human health, influence on the environment and physical and chemical properties.

The legislative framework of PPPs also regulates maximum residue levels (MRLs) of active substances in food and feed commodities. The setting and compliance of MRLs in food and feed commodities are regulated by Regulation (EC) No 396/2005.⁵ MRLs are set at a low level that poses no health risk to consumers in line with good agricultural practices. In addition, other relevant European Legislations and guidance documents are also applicable of PPPs. These include Regulation (EC) No 1272/2008⁶ for the classification and labelling of active substances and PPPs as well as Directive 2009/128/EC⁷ establishing a framework for Community action to achieve the sustainable use of pesticides.

2.3. Risk assessment areas of PPPs and active substance

Risk assessment of active substances and PPPs is performed to evaluate efficacy and foreseeable risks, which may have potentially harmful or unacceptable effects on human, animal and the environment health, respectively. For the determination of the approval of active substance and the authorisation of PPPs, specific areas are assessed. The evaluation of active substances and PPPs generally follows the main principles of the risk assessment process by identifying and characterising the hazard, performing an exposure assessment and characterising the risk.

2.3.1. Identity and physicochemical properties

The identity section contains information regarding the applicant and the manufacturing details as well the details of the chemical composition of the technical active substance as manufactured, manufacturing processes and starting materials. In this section, the specifications regarding the minimum purity of the active substance and the maximum content of any impurity are set. For the PPPs the type of PPP, information on co-formulants, manufacturing process and the full composition of the product are described in detail. From a physicochemical point of view, the following characteristics are evaluated for the active substance, i.e. appearance, melting/boiling point, spectra, vapour pressure, solubility (in water and organic solvents) partition coefficient *n*-octanol/water, dissociation in water, safety properties (oxidising and explosive properties, flammability/flash point, self-heating), surface tension. For a PPP the respective physicochemical properties are required, i.e. appearance, safety properties (oxidising and explosive properties, flammability/flash point, self heating), surface tension and viscosity, acidity/alkalinity and pH value and relative/bulk density. In addition the technical characteristics of the product need to be addressed (e.g. wettability, persistent foaming, suspensibility,

² Regulation (EU) No 283/2013 of 1 March 2013 setting out the data requirements for active substances, in accordance with Regulation (EC) No 1107/2009 of the European Parliament and of the Council concerning the placing of plant protection products on the market.

³ Regulation (EU) No 284/2013 of 1 March 2013 setting out the data requirements for plant protection products, in accordance with Regulation (EC) No 1107/2009 of the European Parliament and of the Council concerning the placing of pl.

⁴ Regulation (EU) No 546/2011 of 10 June 2011 implementing Regulation (EC) No 1107/2009 of the European Parliament and of the Council as regards uniform principles for evaluation and authorisation of plant protection products.

⁵ Regulation (EC) No 396/2005 of the European Parliament and of the Council of 23 February 2005 on maximum residue levels of pesticides in or on food and feed of plant and animal origin and amending Council Directive 91/414/EEC. OJ L 70, 16.3.2005, p. 1–16.

⁶ Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives 67/548/EEC and 1999/45/EC, and amending Regulation (EC) No 1907/2006. OJ L 353, 31.12.2008, p. 1–1355.

⁷ DIRECTIVE 2009/128/EC OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 21 October 2009 establishing a framework for Community action to achieve the sustainable use of pesticides. OJ L 309/71

particle size distribution, etc.). The Commission Communication (2013/C 95/01)⁸ (European Commission, 2013) provides the list of test methods and guidance documents relevant to the implementation of the data requirements Regulation as well as the applicable OECD test guidelines. It is highlighted that besides the safety of the active substance and/or PPP and the technical performance of the PPP, several physicochemical properties are important for the risk assessment in human health and environment (e.g. vapour pressure, water solubility, UV/VIS absorption, partition in *n*-octanol/water).

2.3.2. Methods of analysis

Fully validated and adequately sensitive analytical methods are necessary for the determination of active substance, relevant metabolites and impurities in technical active substance and PPPs and for the determination of pesticide residues. These are generally required for pre-registration and post-registration control and monitoring purposes. Methods of analysis for pesticides covers different substrates and matrices such as plants, plant products, food, animal feed, soil, water and air including both risk assessment and monitoring methods. The training focused on the two relevant guidance documents SANCO/3030/99 (European Commission, 2019b) and SANTE/2020/12830.

2.3.3. Efficacy

Efficacy data are evaluated to assess the nature and extent of benefits that accrue following the use of the PPP on particular crop/pest combination. Submitted data for risk assessment reflect several factors including variability in plant health conditions, climatic differences, the range of agricultural practices, the uniformity of the crops, the mode of application, the type of pest and the type of PPP. Relevant trials on the effectiveness and phytotoxicity should be conducted in accordance with the general and specific standards of the European and Mediterranean Plant Protection Organisation (EPPO). The EPPO standards describe the procedures covering all general aspects of the efficacy evaluation such as analysis and number of trials, minimum effective dose, phytotoxicity/crop safety, effects on succeeding crops or adjacent crops, analysis of resistance, minor uses and climatic considerations. In terms of the efficacy section, extrapolation of results to other crops can be also possible. Such extrapolation for effectiveness and crop safety should follow the available EPPO extrapolation tables on a case-by-case manner and expert judgement.

2.3.4. Residues

Residues are traces of pesticides that can be found in food and feed after treatment with a PPP. Under Regulation (EC) 396/2005, MRLs of these residues are set and a risk assessment is performed as to ensure the safety for all European consumer groups, including the vulnerable groups. Therefore, the setting of MRLs is completed before an authorisation for PPPs is granted. As a general principle, MRLs are the upper legal level of a concentration for a pesticide residue in or on food or feed, based on good agricultural practice (GAP) and the lowest consumer exposure necessary to protect vulnerable consumers. The critical GAP meaning, for example, the maximum number of proposed applications, the shortest interval between applications, the maximum application rate and the shorter preharvest interval is taken into consideration when conducting residue trials. The assessment of residues includes the evaluation of relevant studies such as storage stability studies, nature, and magnitude of residues in plants, livestock, processed commodities and succeeding crops. Data can be extrapolated among crops and tolerances from the intended GAP can be acceptable following specific the residue section, guidelines (SANTE/2019/12752) (European Commission, 2019a).

Following the evaluation of these data, a consumer risk assessment is performed as to ensure that food is safe for consumption.

2.3.5. Mammalian toxicology

In the area of mammalian toxicology, risk assessment includes hazard and exposure assessment. Hazard assessment (identification and characterisation) is carried out considering all available information and data on endpoints in line with data requirements as described in Regulations (EC) No

⁸ Commission Communication in the framework of the implementation of Commission Regulation (EU) No 283/2013 of 1 March 2013 setting out the data requirements for active substances, in accordance with Regulation (EC) No 1107/2009 of the European Parliament and of the Council concerning the placing of plant protection products on the market (2013/C 95/01).

283/2013 or 284/2013 for an active substance or a PPP, respectively. These endpoints may include studies on acute toxicity, irritancy and sensitisation conducted on both the active substance and the product as well as studies on genotoxicity/mutagenicity, carcinogenicity, reproductive toxicity (fertility, development), neurotoxicity, target organ toxicity, endocrine disruption, etc., carried out with the active substance. Additional toxicological information on co-formulants is considered, as available. Most of the toxicity studies are performed using experimental animals, although effort is made in developing alternative approaches for Replacement, Reduction and Refinement of animal use. Mechanistic information provides insight on the mechanism of action and human relevance of observed effects. Where relevant, health-based guidance values (HBGVs), i.e. acceptable daily intake (ADI), (acute) acceptable operator exposure level ((A)AOEL) and acute reference dose (ARfD), are established on the basis of the results of animal studies and considering appropriate uncertainty factors for extrapolation to humans ensuring sufficient protection of vulnerable groups of the population.

The non-dietary exposure assessment is carried out on the active substance of a specific PPP for different population groups, i.e. operators, workers, residents and bystanders, according to the EFSA (2022) guidance document and considering different exposure scenarios depending on the proposed uses of the PPP. Operators, workers, residents and bystanders may be exposed to PPPs either directly through contact with the concentrate, spray dilution, spray drift or dust (via dermal or inhalation routes) or indirectly through contact with drift deposits (dermal or ingestion) or vapour drift arising from volatilisation of deposits. In addition, exposure may be acute (during a single day) or short-term (repeated).

At risk characterisation, the acute and short-term exposure estimates (E) are compared to the relevant HBGVs to conclude if the underlying risk is acceptable ($E < \text{HBGV}$) or not ($E > \text{HBGV}$). In cases of unacceptable risk, different risk mitigation measures may be proposed for consideration in decision making during risk management.

2.3.6. Environmental fate and behaviour in the environment

The environmental fate and behaviour of pesticides session has been divided in two parts. In the first part, how the fate and behaviour of an active substance and its metabolites, in different compartments (soil, surface water, sediment groundwater and air), is addressed according to the data requirements of Regulation (EC) 283/2013, based on appropriate OECD test guidelines. Special attention was given in microbial degradation of the active substance in different compartments. The second part focused on the environmental exposure following the use of a PPP. The relevant entry paths to environmental compartments have been discussed (e.g. precipitation, dry deposition, run-off, drainage, drift, leaching). In addition, mechanisms and factors influencing environmental exposure, such as transport mechanisms, use pattern, substance related factors, and environmental conditions have been considered regarding their impact on the exposure. The estimation of predicted environmental concentrations (PECs) of active substances and their metabolites in the relevant environmental conditions is necessary for the evaluation of the exposure in soil, surface water, groundwater, sediment and air according to Regulation (EC) 284/2013. PEC calculations are estimated taking into consideration the realistic worst-case scenario for active substances and relevant metabolites. A worst-case scenario is considered based on the intended uses. If more than one use is proposed the 'risk envelope' approach in accordance with SANCO/11244/2011 rev.5 is recommended (European Commission, 2011). EU agreed models are available to calculate appropriate PEC in soil, surface water, sediment and groundwater (FOCUS, 1997, 2001, 2014). Relevant PEC calculations have been performed by considering specific active substances and GAP tables for better understanding the rationale behind and the impact on the environmental risk assessment. Moreover, the available and EU acceptable risk mitigation measures according to SANCO/10422/2005 (European Commission, 2007) have been discussed and implemented where necessary.

2.3.7. Ecotoxicology

An ecotoxicological evaluation is carried out to assess the impact on non-target organisms (flora and fauna). An impact on non-target organisms can result from single, prolonged or repeated exposure and can be reversible or irreversible. For the determination of an acceptable or non-acceptable risk, the assessment for PPPs is carried out taking into consideration ecotoxicity studies and data from the fate and behaviour section for soil, surface water and sediment. Following the evaluation of relevant studies, the risk assessment is performed using ecotoxicity endpoints and guidance documents for different categories of non-target organisms. These categories include birds and

mammals, aquatic organisms, soil organisms, bees and other non-target arthropods and non-target terrestrial plants.

For birds and mammals, currently the EFSA (2009) guidance is taken into consideration for the risk assessment. An updated guidance document has been recently published by EFSA (2023), which is not yet legally in force. The effects on mortality reproduction (effects at population level) from exposure of birds and mammals to the pesticide/ active substance are assessed by an acute risk assessment and long-term/reproductive risk assessment, respectively. A tiered approach in risk assessment is followed taking into consideration a screening step of 'indicator species' in terms of the worst-case assumption regarding exposure, a first-tier risk assessment for 'generic focal species' in terms of feeding guilds and BBCH stages and a higher-tier/refined risk assessment of 'focal species' that represents the real species found in a particular crop at a particular time (BBCH stage) to determine realistic exposure estimates.

For aquatic organisms, risk assessment is necessary to compare toxicity levels with exposure and is conducted according to the EFSA PPR Panel (2013) guidance. A tiered assessment is carried out, in which the exposure and in some cases the toxicity (e.g. with the use of microcosm studies) are gradually refined from a worst-case towards a more realistic approach. For aquatics, the risk is estimated on basis of EU active substance data tested on aquatic organism groups and PEC calculations for surface water and sediment for the product uses/application rate. Introduction of risk mitigation measures can take place in order for a safe risk to be demonstrated.

For soil organisms and non-target plants, the risk assessment is carried out according to the SANCO/10329/2002 (European Commission, 2002). The risk assessment for soil organisms is performed on the basis of chronic exposure data to estimate the long-term reproductive toxicity effects taking into consideration also the maximum PEC calculations for soil. Furthermore, the risk assessment is performed for earthworms and other non-target soil meso- and macro-fauna. For the PPPs categorised as non-herbicides only a screening test is performed in order to investigate the phytotoxic effects on the non-target plants outside the treated field. For the PPPs that are categorised as herbicides or plant growth regulators, studies with seedling emergence and vegetative vigour examining the effects on non-target plants are required in order for the risk assessment to be completed. For bees and other non-target arthropods, the risk assessment is carried out according to the SANCO/10329/2002 (European Commission, 2002). For non-target arthropods, exposure toxicity is estimated following a tiered approach starting with glass-plate studies on sensitive indicator species at tier 1 and extended laboratory studies on sensitive and additional indicator species at tier 2. In this case, exposure is calculated for both in-field and off-field area (Candolfi, 2002). Currently for honeybees, the potential acute toxicity is assessed in accordance with the Terrestrial Guidance Document and the potential chronic toxicity for honeybee larvae and worker bee is assessed following the EPPO scheme (2010). Generally for PPPs, the risk assessment in accordance with the EFSA (2013) is currently being performed for illustrative purposes given that it is not yet legally in force.

3. Conclusion

The training programme was developed and implemented by the BPI. The work plan provided training and knowledge in areas of the evaluation of pesticides (PPPs and active substances) according to Regulation (EC) No 1107/2009. The work programme allowed the fellow to get a comprehensive overview into the risk assessment of active substance and PPPs. It also provided the opportunity to develop important skills within critical aspects of the risk assessment framework of different areas of pesticide evaluation. This was done specifically by theoretical sessions and experiencing practical examples of evaluated case studies to identify critical areas of concern. The end result allows for the fellow to efficiently use the gained experience and knowledge to contribute to work involving pesticides in particular PPPs.

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Abbreviations

AAOEL	acute acceptable exposure level
ADI	acceptable daily intake
AOEL	acceptable operator exposure level
ARfD	acute reference dose
BBCH	Biologische Bundesanstalt, Bundessortenamt und Chemische Industrie
BPI	Benaki Phytopathological Institute
EPPO	European and Mediterranean Plant Protection Organisation
GAP	Good Agricultural Practice
HBGV	Health-based guidance levels
MRL	maximum residue level
OECD	Organisation for Economic Cooperation and Development
PEC	predicted environmental concentration
PPP	plant protection product